

[733] (CONCLUDED)
ON A FORMULA OF ELIMINATION.

contain each of them the factor $\theta_1\theta_2$, that is, ∇ ; they, in fact, each of them vanish if $\theta_1 = 0$, and they also vanish if $\theta_2 = 0$; or, by a direct substitution, we have

$$\begin{aligned} XW - Y^2 &= \frac{A^4}{(\theta_1 - \theta_2)^2} \cdot (\theta_1 - \theta_2)^2 \theta_1\theta_2, & &= -A^4\theta_1\theta_2, \\ XW - YZ &= -(\theta_1 - \theta_2)^2 (\theta_1 + \theta_2) \theta_1\theta_2, & &= -A^3\theta_1\theta_2 (\theta_1 + \theta_2), \\ YW - Z^2 &= -(\theta_1 - \theta_2)^2 \theta_1\theta_2 \theta_1\theta_2, & &= -A^2\theta_1\theta_2 \theta_1\theta_2. \end{aligned}$$

Or, what is the same thing, these are $= -4\nabla, 2H\nabla, -C\nabla$, respectively; thus the first equation is

$$\begin{aligned} &\{3A^2c - 6ABb + (-AC + 4B^2)a\} \{2ABd - 3ACc + C^2a\} \\ &\quad - (-A^2d + 3ACb - 2BCa)^2 = -A \cdot (A^3d + \&c.), = -2I\nabla; \end{aligned}$$

and similarly for the other two equations. The nodal curve is thus given by the twofold system $X = 0, Y = 0, Z = 0, W = 0$.

The method may be extended to the case where, instead of the quadric equation $(A, B, C \mid \vartheta, 1)^2 = 0$, we have an equation of any higher order, but the formulae are less simple.

734.

ON THE KINEMATICS OF A PLANE.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. xvi. (1879), pp. 1–8.]

It seems desirable to bring together under this title various questions which have been, or may be, proposed or discussed. We consider two planes in relative motion one upon the other, but, for convenience, they may be distinguished as a moving plane and a fixed plane, the first moving upon the second. Any point of the moving plane traces out on the fixed plane a curve, and any line of the moving plane envelopes on the fixed plane a curve; similarly, any point of the fixed plane traces out on the moving plane a curve, and any line of the fixed plane envelopes on the moving plane a curve. More generally, any curve of the moving plane envelopes on the fixed plane a curve, and any curve of the fixed plane envelopes on the moving plane a curve. There is, moreover, in the moving plane a curve which rolls upon a curve in the fixed plane, and these two curves (a single relative position being given) determine the motion.

[Figure 1: Fixed axes Ox, Oy , and moveable axes O_1x_1, O_1y_1 , with the origin O_1 at position (α, β) and O_1x_1 inclined at angle θ to Ox ; a point P shown with respect to both sets of axes.]

The analytical theory presents no difficulty. Taking in the fixed plane the fixed axes Ox, Oy (fig. 1), and, fixed in the moveable plane so as to move with it, the axes O_1x_1, O_1y_1 ; then the position of the axes $O_1x_1y_1$ may be determined, say by α, β , the coordinates of O_1 in regard to Oxy ; and by θ , the inclination of O_1x_1 to Ox . And denoting by x, y, x_1, y_1 the coordinates of a point P in regard to the two sets of axes respectively, then

$$\begin{aligned} x &= \alpha + x_1 \cos \theta - y_1 \sin \theta, \\ y &= \beta + x_1 \sin \theta + y_1 \cos \theta; \end{aligned}$$

or, what is the same thing,

$$\begin{aligned} x_1 &= (x - \alpha) \cos \theta + (y - \beta) \sin \theta, \\ y_1 &= -(x - \alpha) \sin \theta + (y - \beta) \cos \theta; \end{aligned}$$

or, as these last equations may be written,

$$\begin{aligned}x_1 &= \alpha_1 + x \cos(-\theta) - y \sin(-\theta), \\y_1 &= \beta_1 + x \sin(-\theta) + y \cos(-\theta),\end{aligned}$$

where $\alpha_1, \beta_1, = -\alpha \cos \theta - \beta \sin \theta, \alpha \sin \theta - \beta \cos \theta$, are the coordinates of O referred to the axes $O_1x_1y_1$, and $-\theta$ is the inclination of Ox to O_1x_1 .

When the motion is given, α, β, θ are given functions of a single variable parameter, say of t^* ; or, if we please, α, β are given functions of θ .

The velocities of a given point (x, y) are determined by the equations

$$\begin{aligned}x' &= \alpha' - (x_1 \sin \theta + y_1 \cos \theta) \theta', \\y' &= \beta' + (x_1 \cos \theta - y_1 \sin \theta) \theta';\end{aligned}$$

that is,

$$\begin{aligned}x' - \alpha' &= -(y - \beta) \theta', \\y' - \beta' &= (x - \alpha) \theta';\end{aligned}$$

or, as these equations may also be written,

$$\begin{aligned}(x' - \alpha') \sin \theta - (y' - \beta') \cos \theta &= -x_1 \theta', \\-(x' - \alpha') \cos \theta - (y' - \beta') \sin \theta &= y_1 \theta' .\end{aligned}$$

Hence if $x' = 0, y' = 0$, we have

$$\begin{aligned}x_1 \theta' &= \alpha' \sin \theta - \beta' \cos \theta, & x_1 &= -(y - \beta) + \frac{\alpha'}{\theta'}, \\y_1 \theta' &= \alpha' \cos \theta + \beta' \sin \theta, & y_1 &= (x - \alpha) + \frac{\beta'}{\theta'},\end{aligned}$$

which equations determine in terms of t, x_1 and y_1 the coordinates in regard to the axes $O_1x_1y_1$, and x and y the coordinates in regard to the axes Oxy , of I , the centre of instantaneous rotation.

If from the expressions of x_1, y_1 we eliminate t , we obtain an equation between (x_1, y_1) , which is that of the rolling curve in the moveable plane; and, similarly, if

from the expressions of x, y we eliminate t , we obtain a relation between (x, y) , which is that of the rolled-on curve in the fixed plane.

The system may be written

$$\begin{aligned}x_1 &= -\frac{\alpha'}{\theta'} \sin \theta + \frac{\beta'}{\theta'} \cos \theta, & x &= \alpha - \frac{\beta'}{\theta'}, \\y_1 &= \frac{\alpha'}{\theta'} \cos \theta + \frac{\beta'}{\theta'} \sin \theta, & y &= \beta + \frac{\alpha'}{\theta'};\end{aligned}$$

or, if we take θ as the independent variable,

$$\begin{aligned}x_1 &= -\alpha' \sin \theta + \beta' \cos \theta, & x &= \alpha - \beta', \\y_1 &= \alpha' \cos \theta + \beta' \sin \theta, & y &= \beta + \alpha' .\end{aligned}$$

To find the variations of I , we have

$$\begin{aligned}x'_1 &= \alpha'' \sin \theta - \beta'' \cos \theta + \alpha' \cos \theta + \beta' \sin \theta, & &= \alpha'' \sin \theta - \beta'' \cos \theta + y_1, \\y'_1 &= \alpha'' \cos \theta + \beta'' \sin \theta - \alpha' \sin \theta + \beta' \cos \theta, & &= \alpha'' \cos \theta + \beta'' \sin \theta - x_1, \\x' &= \alpha' - \beta'', \\y' &= \beta' + \alpha'' .\end{aligned}$$

^{0*} t may be regarded as denoting the time, and then the derived functions of x, y in regard to t will denote velocities; and, to simplify the expression of the theorems, it is convenient to do this.

Hence

$$\begin{aligned} x'_1 &= x' \cos \theta + y' \sin \theta, & \text{or } x' &= x'_1 \cos \theta - y'_1 \sin \theta, \\ y'_1 &= -x' \sin \theta + y' \cos \theta, & \text{or } y' &= x'_1 \sin \theta + y'_1 \cos \theta, \end{aligned}$$

values which give $x'^2 + y'^2 = x_1'^2 + y_1'^2$, which equation expresses that the motion is in fact a rolling one.

Imagine the two curves, and the initial relative position given; say the two points A, A_1 (fig. 2) were originally in contact, then the arcs AI, A_1I are equal, and, calling each of these s , and X, Y, X_1, Y_1 the coordinates of I in regard to the two

[Figure 2: Two curves in contact at I ; the fixed-plane curve runs through A and the moveable-plane curve through A_1 , with arc-lengths $AI = A_1I = s$.]

sets of axes respectively, we have X, Y, X_1, Y_1 given functions of s , such that $X'^2 + Y'^2 = 1, X_1'^2 + Y_1'^2 = 1$, the accents now denoting differentiation in regard to s . We have, from the figure,

$$\theta = \tan^{-1} \frac{Y'}{X'} - \tan^{-1} \frac{Y'_1}{X'_1};$$

or, what is the same thing,

$$\tan \theta = (Y'X'_1 - Y'_1X') \div (X'X'_1 + Y'Y'_1),$$

say

$$\sin \theta, \cos \theta = Y'X'_1 - Y'_1X', \quad X'X'_1 + Y'Y'_1;$$

and then, as before,

$$\begin{aligned} x &= \alpha + x_1 \cos \theta - y_1 \sin \theta, \\ y &= \beta + x_1 \sin \theta + y_1 \cos \theta; \end{aligned}$$

or, what is the same thing,

$$\begin{aligned} x - X &= \cos \theta (x_1 - X_1) - \sin \theta (y_1 - Y_1), \\ y - Y &= \sin \theta (x_1 - X_1) + \cos \theta (y_1 - Y_1), \end{aligned}$$

where X, Y, X_1, Y_1 , and therefore also θ , denote given functions of s . The formulae will be of a like form if X, Y, X_1, Y_1 are given functions of a parameter t .

A well known but very interesting case is when two points of the moving plane describe right lines on the fixed plane. This may be discussed geometrically as follows: Suppose that we have the points A, C (fig. 3) describing the lines OA_0, OC_0 , which meet in O ; through A, C, O describe a circle, centre O_1 , and with centre

[Figure 3: Two perpendicular fixed lines OA_0, OC_0 meeting at O ; small circle through A, C, O with centre O_1 rolls inside large circle of radius $= 2OO_1$ through O ; tangency point I , with A_0, C_0 on the large circle.]

O and radius $= 2OO_1$, describe a circle touching the first circle in a point I ; and suppose that A_0, C_0 denote points on the second circle. Then it is at once seen that, considering the first or small circle as belonging to the moving plane, and the second or large circle as belonging to the fixed plane, the motion is in fact the rolling motion of the small upon the large circle; and, moreover, that each point of the small circle describes a right line, which is a diameter of the large circle. In fact, the angle IO_1C at the centre is the double of the angle IOC at the circumference; that is, it is the double of the angle IOC_0 ; and therefore (the radius of the small circle being half that of the large circle) the arcs IC, IC_0 are equal, so that the rolling motion will carry the point C along the radius OC_0 , and will, in like manner, carry the point A along the radius OA_0 , or the motion will be as originally assumed. And, in like manner, for any other point B of the small circle the motion will be along the radius OB_0 ; in particular, taking AB a diameter, the angle A_0OB_0 will be a right angle; and the motion is determined by means of the two points A, B describing respectively the two lines OA_0, OB_0 at right angles to each other, viz. there is no loss of

generality in assuming that the two fixed lines are at right angles to each other. It thence at once follows, as will presently appear, that each point of the moving plane describes an ellipse (but we have the special case already referred to, each point on the small circle describes a right line, and also the special case, the centre O_1 of the small circle describes a circle). Considering any point Q of the moving plane, let the line QO_1 meet the small circle in the points E, F (or, what is the same thing, let E, F be the extremities of the diameter which passes through Q); then the points E, F describe the lines OE, OF at right angles to each other, and Q is a point on EF or on this line produced; clearly the locus is an ellipse having the lines OE, OF for the directions of its axes, and having the lengths of the semi-axes $= QF, QE$ respectively.

Taking the points to be A, B moving along the two lines OB_0, OA_0 at right angles to each other, these lines may be taken for the axes Ox, Oy ; the point O_1 for the origin of the coordinates x_1, y_1 , the axes O_1x_1 being in the direction O_1B and O_1y_1 at right angles to it; calling the length $AB = 2c$, we have $O_1A = O_1B = c$, and the angle ABO may be called θ (but this angle was previously taken with a contrary sign). We have then for the point P , having in regard to O_1x_1 and O_1y_1 the coordinates (x_1, y_1) ,

$$\begin{aligned} x &= \alpha + x_1 \cos \theta + y_1 \sin \theta, \\ y &= \beta - x_1 \sin \theta - y_1 \cos \theta, \end{aligned}$$

where the sign of y_1 has been changed, and $\alpha = c \cos \theta, \beta = c \sin \theta$: the equations thus become

$$\begin{aligned} x &= (c + x_1) \cos \theta + y_1 \sin \theta, \\ y &= (c - x_1) \sin \theta - y_1 \cos \theta, \end{aligned}$$

where observe that $c + x_1, c - x_1$ are the distances M_1A, M_1B respectively. And we have, conversely,

$$\begin{aligned} x_1 &= x \cos \theta - y \sin \theta - c \cos 2\theta, \\ y_1 &= -x \sin \theta - y \cos \theta + c \sin 2\theta. \end{aligned}$$

If, in particular, $y_1 = 0$, then

$$x = (c + x_1) \cos \theta, \quad y = (c - x_1) \sin \theta;$$

or we have

$$\frac{x^2}{(c + x_1)^2} + \frac{y^2}{(c - x_1)^2} = 1,$$

viz. the curve on the first plane is an ellipse, the semi-axes of which are $\pm(c + x_1), \pm(c - x_1)$, each taken positively; if $x_1^2 + y_1^2 = c^2$, viz. if P be on the circle having AB for its diameter, then $y_1^2 = (c + x_1)(c - x_1)$, and we have

$$\frac{y}{x} = \frac{-(c - x_1) \cos \theta + y_1 \sin \theta}{(c + x_1) \sin \theta + y_1 \cos \theta}, \quad = -(c - x_1) \div y_1,$$

viz. as mentioned above, the curve on the fixed plane is a right line.

In the general case, we have

$$\begin{aligned} x(c - x_1) + yy_1 &= -(c^2 - x_1^2 - y_1^2) \cos \theta, \\ xy_1 + y(c + x_1) &= -(c^2 - x_1^2 - y_1^2) \sin \theta, \\ \{x(c - x_1) + yy_1\}^2 + \{xy_1 + y(c + x_1)\}^2 &= (c^2 - x_1^2 - y_1^2)^2, \end{aligned}$$

or, what is the same thing,

$$x^2\{(c - x_1)^2 + y_1^2\} + 4xy_1cy_1 + y^2\{(c + x_1)^2 + y_1^2\} = (c^2 - x_1^2 - y_1^2)^2.$$

Considering (x_1, y_1) as given, the curve traced out by P on the fixed plane is of the second order; it would be easy to verify from the equation that it is an ellipse, and to obtain for the position and magnitude of the axes the construction already found geometrically.

The same equation, considering therein (x, y) as constant and (x_1, y_1) as current coordinates, gives the curve traced out on the moving plane; the curve is obviously of the fourth order. Transferring the origin to A , we must in place of x_1 write $x_1 - c$; the equation thus becomes

$$x^2\{(x_1 - 2c)^2 + y_1^2\} + 4xy_1xy + y^2(x_1^2 + y_1^2) = (x_1^2 + y_1^2 - 2cx_1)^2,$$

or, what is the same thing,

$$(x_1^2 + y_1^2 - 2cx_1)^2 = (x^2 + y^2)(x_1^2 + y_1^2) - 4cx(xx_1 + yy_1) + 4c^2x^2;$$

and if we suppose herein $x = 0$, it becomes

$$(x_1^2 + y_1^2 - 2cx_1)^2 = (x_1^2 + y_1^2)y^2;$$

or, writing $x_1 = r_1 \cos \theta_1$, $y_1 = r_1 \sin \theta_1$, where $\theta_1 = \text{angle } QAB$, this is

$$(r_1 - 2c \cos \theta_1)^2 - y^2 = 0,$$

or say it is

$$r_1 = 2c \cos \theta_1 \pm y,$$

which is the polar equation of the curve described on the moveable plane by the point S , whose coordinates in respect to Ox and Oy are $(0, y)$.

There is no loss of generality in assuming $x = 0$. In fact, starting with any point S whatever of the fixed plane, if we draw OS meeting the small circle in A , and through A draw at right angles to this a line meeting the same circle in B , then, as before, the points A and B move along the fixed lines OA_0 , OB_0 ; or as regards the relative motion, taking A, B as fixed points, we have the originally fixed plane now moving in such wise that the two lines OA_0 , OB_0 thereof (at right angles to each other) pass always through the points A and B respectively, and the curve is that described by the point S on the line OA ; the point O describes the circle on the diameter AB (the small circle), equation $r_1 = 2c \cos \theta_1$; and OQ having a given constant value $= y$, we have for the curve described by the point S the foregoing equation $r_1 = 2c \cos \theta_1 - y$; or writing $y = -f$, that is, taking S on the other side of O at a distance $OS = f$, the equation is $r_1 = 2c \cos \theta_1 + f$; viz. this is a nodal Cartesian or Limaçon, the origin being an acnode or a crunode according as $f >$ or $< 2c$; and if $f = 2c$, then we have the cuspidal curve or cardioid $r_1 = 2c(1 + \cos \theta_1) = 4c \cos^2 \frac{1}{2}\theta_1$.

The general conclusion is that the centre O of the large circle describes on the moving plane a small circle (centre O_1), and that every other point of the fixed plane describes on the moving plane a Limaçon having for its node a point of the small circle, and being, in fact, the curve obtained by measuring off along the radius vector of the small circle from its extremity a constant distance.

Considering in connexion with the point, coordinates (x_1, y_1) , (x, y) , a second point, coordinates (X_1, Y_1) , (X, Y) , in regard to the two sets of axes respectively, we have

$$\begin{aligned} x &= (c + x_1) \cos \theta - y_1 \sin \theta, & X &= (c + X_1) \cos \theta - Y_1 \sin \theta, \\ y &= (c - x_1) \sin \theta - y_1 \cos \theta, & Y &= (c - X_1) \sin \theta - Y_1 \cos \theta; \end{aligned}$$

from the first two equations we have

$$\cos \theta : \sin \theta : 1 = x(c - x_1) + yy_1 : xy_1 + y(c + x_1) : c^2 - x_1^2 - y_1^2,$$

and substituting these values in the second set, we find

$$\begin{aligned} X : Y : 1 \\ &= x[c^2 - c(X_1 - x_1) - X_1x_1 - Y_1y_1] + y[c(y_1 - Y_1) - y_1X_1 - x_1Y_1] \\ &: x[c(y_1 - Y_1) - y_1X_1 + x_1Y_1] + y[c^2 - c(X_1 - x_1) - X_1x_1 - Y_1y_1] \\ &: c^2 - x_1^2 - y_1^2; \end{aligned}$$

or the points (x, y) , (X, Y) , considered as each of them moving on the fixed plane, are homographically related to each other.

To find the curve enveloped on the fixed plane by a given curve of the moving plane, we have only in the equation $f(x_1, y_1) = 0$ of the curve in the moving plane to substitute for x_1, y_1 their values in terms of x, y, θ , and then considering θ as a variable parameter, to find the envelope of the curve represented

by this equation. And, similarly, we find the curve enveloped on the moving plane by a given curve of the fixed plane.

Thus, in the particular case of motion above considered, writing, as before,

$$\begin{aligned}x &= (c + x_1) \cos \theta - y_1 \sin \theta, \\y &= (c - x_1) \sin \theta - y_1 \cos \theta;\end{aligned}$$

or conversely

$$\begin{aligned}x_1 &= x \cos \theta - y \sin \theta - c \cos 2\theta, \\y_1 &= -x \sin \theta - y \cos \theta + c \sin 2\theta;\end{aligned}$$

the envelope on the moving plane of the line

$$Ax + By + C = 0$$

of the fixed plane is given as the envelope of the line

$$\{A(c + x_1) - By_1\} \cos \theta + \{-Ay_1 + B(c - x_1)\} \sin \theta + C = 0;$$

that is,

$$\{A(c + x_1) - By_1\}^2 + \{Ay_1 - B(c - x_1)\}^2 - C^2 = 0,$$

viz. this is

$$(A^2 + B^2)(x_1^2 + y_1^2 + c^2) + 2(A^2 - B^2)cx_1 - 4ABcy_1 - C^2 = 0,$$

a circle.

But the envelope on the fixed plane of the line

$$Ax_1 + By_1 + C = 0$$

of the moving plane is given as the envelope of the line

$$C + (Ax + By) \cos \theta - (Ay + Bx) \sin \theta - AC \cos 2\theta + BC \sin 2\theta = 0,$$

which can be obtained by equating to zero the discriminant of a quartic function, and is apparently a sextic curve.

735.

NOTE ON THE THEORY OF APSIDAL SURFACES.

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I OBTAIN in the present Note a system of formulae which lead very simply to the known theorem, that the apsidals of reciprocal surfaces are reciprocal; or, what is the same thing, that the reciprocal of the apsidal of a given surface is the apsidal of its reciprocal; the surfaces are referred to the same axes, and by the reciprocal is meant the reciprocal surface in regard to a sphere radius unity, having for its centre a determinate point, say the origin; and it is this same point which is used in the construction of the apsidal surfaces. The apsidal of a given surface is constructed as follows; considering the section by any plane through the fixed point, and in this section the apsidal radii from the fixed point (that is, the radii which meet the curve at right angles), then drawing a line through the fixed point at right angles to the plane, and on this line measuring off from the fixed point distances equal to the apsidal radii respectively, the locus of the extremities of these distances is the apsidal surface. We have the surface, its reciprocal, the apsidal of the surface, the apsidal of the reciprocal; and I take

$$(x, y, z), (x', y', z'), (X, Y, Z), (X', Y', Z')$$

for the coordinates of corresponding points on the four surfaces respectively.

The condition of reciprocity gives $xx' + yy' + zz' - 1 = 0$, and (the equations being $U = 0$, $U' = 0$) x' , y' , z' proportional to $d_x U$, $d_y U$, $d_z U$, and x , y , z proportional to $d_x U'$, $d_y U'$, $d_z U'$; or, what is the same thing, we must have

$$x' dx + y' dy + z' dz = 0 \quad \text{and} \quad x dx' + y dy' + z dz' = 0;$$

one of these is implied in the other, as appears at once by differentiating the equation $xx' + yy' + zz' - 1 = 0$.

The other two surfaces will therefore be reciprocal if only we have the like relations between the coordinates (X, Y, Z) and (X', Y', Z') ; that is, if

$$\begin{aligned} XX' + YY' + ZZ' - 1 &= 0, \\ X' dX + Y' dY + Z' dZ &= 0, \\ X dX' + Y dY' + Z dZ' &= 0. \end{aligned}$$

To find the apsidal surface, we consider an arbitrary section $x \cos \alpha + y \cos \beta + z \cos \gamma = 0$ of the surface $U = 0$, and seek to determine the apsidal radii thereof, that is, the maximum or minimum values of $R^2 = x^2 + y^2 + z^2$ when x , y , z vary subject to these two conditions. Writing x' , y' , z' to denote functions proportional to $d_x U$, $d_y U$, $d_z U$, we thus have the set of equations

$$\begin{aligned} x + \lambda x' + \mu \cos \alpha &= 0, \\ y + \lambda y' + \mu \cos \beta &= 0, \\ z + \lambda z' + \mu \cos \gamma &= 0, \end{aligned}$$

where λ , μ are indeterminate coefficients; taking then X , Y , Z as the coordinates of the extremity of the line drawn at right angles to the plane, we have $R^2 = X^2 + Y^2 + Z^2$, and $\cos \alpha$, $\cos \beta$, $\cos \gamma = \frac{X}{R}$, $\frac{Y}{R}$, $\frac{Z}{R}$; substituting these values in the equation

$$x \cos \alpha + y \cos \beta + z \cos \gamma = 0,$$

we have $Xx + Yy + Zz = 0$, and substituting in the other equations, and instead of λ , μ introducing the new indeterminate coefficients ρ , σ , we obtain

$$X, Y, Z = \rho x + \sigma x', \quad \rho y + \sigma y', \quad \rho z + \sigma z'.$$

Hence these last equations, together with

$$R^2 = X^2 + Y^2 + Z^2 = x^2 + y^2 + z^2,$$

and

$$Xx + Yy + Zz = 1,$$

contain the solution of the problem. If for convenience we introduce R'^2 to denote $x'^2 + y'^2 + z'^2$, and imagine the absolute values of x' , y' , z' determined so that $xx' + yy' + zz' = 1$, then substituting for X , Y , Z their values in the equations $X^2 + Y^2 + Z^2 = R^2$ and $Xx + Yy + Zz = 1$, we find

$$R^2 = \rho^2 R^2 + 2\rho\sigma + \sigma^2 R'^2, \quad 1 = \rho R^2 + \sigma,$$

and thence

$$\rho^2 = \frac{1}{R^2 R'^2 - 1}, \quad \sigma = -\rho R^2,$$

or, finally assuming

$$\rho = \frac{1}{\sqrt{R^2 R'^2 - 1}},$$

we have

$$X, Y, Z = x - R^2 x', \quad y - R^2 y', \quad z - R^2 z',$$

each divided by

$$\sqrt{R^2 R'^2 - 1},$$

where I recall that x', y', z' are proportional to $d_x U, d_y U, d_z U$, and are such that $xx' + yy' + zz' = 1$: they in fact denote

$$d_x U, d_y U, d_z U, \text{ each divided by } x d_x U + y d_y U + z d_z U,$$

and that R^2 and R'^2 denote $x^2 + y^2 + z^2$ and $x'^2 + y'^2 + z'^2$ respectively. The coordinates X, Y, Z of the point of the apsidal surface are thus determined as functions of x, y, z .

For the apsidal of the reciprocal surface, we have in like manner

$$X', Y', Z' = x' - R'^2 x, y' - R'^2 y, z' - R'^2 z,$$

each divided by

$$\sqrt{R^2 R'^2 - 1},$$

and then the two sets of values give, not only

$$XX' + YY' + ZZ' = 1,$$

as is obvious, but also

$$X' dX + Y' dY + Z' dZ = 0, \quad \text{and} \quad X dX' + Y dY' + Z dZ' = 0.$$

In fact, writing for a moment p, p' instead of R^2, R'^2 , and $\sqrt{R^2 R'^2 - 1} = \sqrt{pp' - 1} = \omega$, then

$$\begin{aligned} X' dX + Y' dY + Z' dZ &= \sum \frac{x' - xp'}{\omega} d\left(\frac{x - x'p}{\omega}\right) + \&c. \\ &= \sum \frac{x' - xp'}{\omega} \left\{ \frac{dx - p dx' - x' dp}{\omega} - \frac{(x - x'p) d\omega}{\omega^2} \right\} + \&c. \\ &= \frac{1}{\omega^2} \left\{ \begin{aligned} &x' dx + y' dy + z' dz \\ &- p (x' dx' + y' dy' + z' dz') \\ &- (x'^2 + y'^2 + z'^2) dp \\ &- p' (x dx + y dy + z dz) \\ &+ pp' (x dx' + y dy' + z dz') \\ &+ p' (xx' + yy' + zz') dp \end{aligned} \right\} \\ &\quad - \frac{d\omega}{\omega^3} \left\{ \begin{aligned} &xx' + yy' + zz' \\ &- p (x'^2 + y'^2 + z'^2) \\ &- p' (x^2 + y^2 + z^2) \\ &+ pp' (xx' + yy' + zz') \end{aligned} \right\}, \end{aligned}$$

or, since the terms in $\{ \}$ are

$$0 - p \cdot \frac{1}{2} dp' - p' dp - p' \cdot \frac{1}{2} dp + 0 + p' dp = -\frac{1}{2}(p dp' + p' dp),$$

and

$$1 - pp' - pp' + pp' = 1 - pp' = -\omega^2,$$

this is

$$= \frac{1}{\omega^2} \left\{ -\frac{1}{2}(p dp' + p' dp) + \omega d\omega \right\} = 0,$$

in virtue of $\omega^2 = pp' - 1$. And similarly the other equation $X dX' + Y dY' + Z dZ' = 0$ might be directly verified.

736.

APPLICATION OF THE NEWTON-FOURIER METHOD TO AN
IMAGINARY ROOT OF AN EQUATION.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. xvi. (1879),
pp. 179–185.]

I CONSIDER only the most simple case, that of a quadric equation $x^2 = n^2$, where n^2 is a given imaginary quantity, having the square roots n and $-n$; starting from an assumed approximate (imaginary) value $x = a$, we have $(a + h)^2 = n^2$, that is,

$$a^2 + 2ah = n^2, \quad h = \frac{n^2 - a^2}{2a}, \quad \text{and} \quad a + h = \frac{a^2 + n^2}{2a},$$

that is, the successive values are

$$a_1 = \frac{a^2 + n^2}{2a}, \quad a_2 = \frac{a_1^2 + n^2}{2a_1}, \quad \dots,$$

and the question is, under what conditions do we thus approximate to one determinate root (selected out of the two roots at pleasure), say n , of the given equation.

The nearness of two values is measured by the modulus of their difference; thus a nearer to n , than a_1 is to n , means $\text{mod}(a - n) < \text{mod}(a_1 - n)$, and so in other cases; in the course of the approximation $a, a_1, a_2, \dots$ to n , any step, for instance a to a_1 , is regular if a_1 is nearer to n than a is, but otherwise it is irregular; the approximation is regular if all the steps are regular, and if (after one or more irregular steps) all the subsequent steps are regular, then the approximation becomes regular at the step which is the first of the unbroken series of regular steps.

We do by an approximation, which is ultimately regular, obtain the value n , if only the assumed value a is nearer to n than it is to $-n$; or, say, if the condition $\text{mod}(a - n) < \text{mod}(a + n)$ is satisfied, and the approximation is regular from the beginning if $\text{mod}(a - n) \leq \frac{1}{2}\text{mod}n$, viz. this condition is a sufficient one¹; the first step a to a_1 will moreover be regular under a less stringent condition imposed upon a ; and it would seem that, without the condition $\text{mod}(a - n) < \frac{1}{2}\text{mod}n$ being satisfied, the subsequent steps will in some cases be also regular; that is, that the last-mentioned condition is not a necessary condition in order to the approximation being regular from the beginning; it is, however, the necessary and sufficient condition, to be satisfied by the modulus of $a - n$, in order that the approximation may be regular from the beginning. All this will clearly appear from the geometry.

[Figure 1: Points N, N' on a horizontal axis with O at the midpoint; through O a perpendicular QQ' . A circle on $N'M$ as diameter where $OM = \frac{1}{2}ON$ (the “circle of unfitness”); the segment on the N -side of QQ' is the “segment of unfitness.” Points $A, A_1, \dots$ representing successive approximations.]

We take N, N' (fig. 1) to represent the values $n, -n$; and similarly $A, A_1, \&c.$ to represent the quantities $a, a_1, \dots$; we have then

$$AN = \text{mod}(a - n), \quad A_1N = \text{mod}(a_1 - n) \dots,$$

so that the approximation is measured by the approach of the points A, A_1 to N . The line NN' joining the points N, N' passes through, and is bisected at, the origin O ; drawing then QQ' through O at right angles to NN' the condition

$$\text{mod}(a - n) < \text{mod}(a + n)$$

means that the point A , which represents the imaginary quantity a , lies on the N -side of QQ' , and it will be assumed throughout that this is so. Take now on the line ON , $OM = \frac{1}{2}ON$, and on $N'M$ as diameter, describe a circle, which may be called the “circle of unfitness”; regarding as an area the segment hereof

¹In the Smith's Prize Examination, Jan. 28, 1879, I gave the theorem under the following form: “If a, n are imaginary quantities, the latter of them given, and the former assumed at pleasure, subject only to the condition $\text{mod}(a - n) < \frac{1}{2}\text{mod}n$; then if $a_1 = \frac{a^2 + n^2}{2a}$, $a_2 = \frac{a_1^2 + n^2}{2a_1}$, &c., show that the terms $a, a_1, a_2, \dots$ will converge to the limit n .” This is strictly true, but it would have been better to say “will converge regularly.”

which lies on the N -side of QQ' , say this is the “segment of unfitness.” It will be shown that according as A is situate inside, on the boundary of, or outside the segment of unfitness, A_1N will be greater than, equal to, or less than AN . It may be added that, if A be within or upon the boundary of the segment of unfitness, then A_1 will be outside it, but this by no means hinders that the next point A_2 , or some later point, shall be within the segment of unfitness; and, further, that when A is outside the segment of unfitness, then the next point A_1 , or some later point, may very well be within the segment of unfitness; the conclusion is, that A being inside the segment of unfitness, A_1N is less than AN , but that it does not thence follow that A_2N is less than A_1N , A_3N than A_2N , ...; the approximation although regular at the first step, may then, or afterwards, for a step or steps, cease to be regular.

If, however, AN be less than $\frac{1}{2}ON$, that is, if the condition $\text{mod}(a - n) < \frac{1}{2}\text{mod}n$ be satisfied, then the point A lies within the circle centre N and radius NM , and is consequently outside the segment of unfitness; A_1N being less than AN , the point A_1 is a fortiori outside the segment of unfitness, and the like for all the subsequent points $A_2, A_3, \dots$; that is, in this case, the approximation is regular throughout. The circle, centre N , and radius $NM, = \frac{1}{2}\text{mod}n$, may be called the “safe circle”; and the conclusion is that, if the point A or any subsequent point be within the safe circle, then every subsequent point will be within the safe circle, and the approximation will be regular.

The successive points $A, A_1, A_2, \dots$ (or, as it will be convenient to call them, $A_1, A_2, A_3, \dots$) may be obtained each from the preceding one by a simple geometrical construction.

I recall that any circle through the two (imaginary) antipoints of N, N' is a circle having its centre on the indefinite line NN' ; it is such that the ratio of the distances of a point thereof from the points N, N' respectively has a certain constant value, viz. for the circles with which we are here alone concerned, those which lie on the N -side of QQ' , the centres lie beyond the point N (further away, that is, from O), and the values of the ratio, distance from N to distance from N' , are less than unity.

Starting then from the given point A_1 , for which this ratio $A_1N : A_1N'$ has a given value, suppose $A_1N = k A_1N'$, we describe a first circle (passing of course through A_1) for each point of which this ratio has the value k ; let the diameter of this circle be V_1W_1 , V_1 being the extremity between O and N , W_1 (not shown in the figure), that beyond N ; we then describe a second circle, for which the ratio is $= k^2$; let its diameter be V_2W_2 , V_2 being the extremity between N and O (or say between V_1 and N), W_2 , that beyond N (or say between N and W_1); the point A_2 lies on this second circle, and is determined as the single intersection of the line V_2A_1 with the second circle. And of course drawing a third circle, for which the ratio is $= k^4$, on the diameter V_3W_3 , then A_3 lies on the third circle, and is the intersection with it of the line V_3A_2 , and so on; the radii of the successive circles diminish very rapidly, their centres, in like manner, continually approaching the point N ; hence, the points $A_1, A_2, A_3, \dots$, which lie on the several circles respectively approximate, and that very rapidly, to the point O . But by what precedes, if, for instance, the point A_1 be within the segment of unfitness, then also some of the subsequent points may be within the segment of unfitness, and for each point A_p , for which this is the case, the next point A_{p+1} is at a greater distance, so that $NA_{p+1} > NA_p$; it is, however, clear that we always arrive at a point A_q , such that $NA_q < \frac{1}{2}ON$, and so soon as such a point is arrived at the approximation becomes regular.

The point A_2 determined from A_1 , as above, is a point such that the subtended angle NA_2N' is $=$ twice the subtended angle NA_1N' ; or calling the latter angle ϕ , the former is 2ϕ . It is, in fact, this property which gives rise to the construction; for let the values of A_1N, A_1N' , regarded as imaginary quantities, be called for a moment

$$\rho_1(\cos \theta_1 + i \sin \theta_1), \quad \rho'_1(\cos \theta'_1 + i \sin \theta'_1);$$

and, similarly, those of A_2N, A_2N' be called

$$\rho_2(\cos \theta_2 + i \sin \theta_2), \quad \rho'_2(\cos \theta'_2 + i \sin \theta'_2);$$

then these are the values of $a_1 + n, a_1 - n, a_2 + n, a_2 - n$ respectively, or we have

$$\begin{aligned} \frac{a_1 - n}{a_1 + n} &= \frac{\rho_1}{\rho'_1} [\cos(\theta_1 - \theta'_1) + i \sin(\theta_1 - \theta'_1)] = k (\cos \phi + i \sin \phi), \\ \frac{a_2 - n}{a_2 + n} &= \frac{\rho_2}{\rho'_2} [\cos(\theta_2 - \theta'_2) + i \sin(\theta_2 - \theta'_2)] = k^2 (\cos 2\phi + i \sin 2\phi), \end{aligned}$$

that is,

$$\left(\frac{a_1 - n}{a_1 + n} \right)^2 = \frac{a_2 - n}{a_2 + n},$$

which relation between a_2, a_1 is in fact the original relation

$$a_2 = \frac{a_1^2 + n^2}{2a_1};$$

and, conversely, a_2, a_1 , being thus connected, then the representative A_2 is obtained from the representative point A_1 by the foregoing geometrical construction.

I give the analytical proofs; we may without loss of generality take, and it is convenient to do so, the axis of x as coinciding with the line ON , and to put also $ON = 1$. We then in place of the original coordinates x, y of any point take the new coordinates k, ϕ , which are such that

$$\frac{x^2 + y^2 - 1 + 2iy}{x^2 + y^2 + 1 - 2x} = k e^{i\phi},$$

$$\frac{x - iy - 1}{x - iy + 1} = k e^{-i\phi},$$

equations which may also be written

$$(x - iy + 1)^2 = e^{i\phi} [(x + iy + 1)^2],$$

or, what is the same thing,

$$\begin{aligned} x^2 + (y - i)^2 &= e^{-2i\phi} [x^2 + (y + i)^2]; \\ x^2 + y^2 + 2y \cot \phi &= 0, \end{aligned}$$

where of course the equation with k shows that k is equal to the ratio of the distances of the point from the points N, N' respectively, and the equation in ϕ , taken in the second form, shows that ϕ is the angle subtended at the point by N, N' .

It is sometimes convenient to write $ke^{i\phi}, ke^{-i\phi} = p, q$ respectively; we then have

$$x = \frac{p+q}{1-p} \cdot \frac{1}{1-q} \cdot p, \quad x - iy = \frac{1+p}{1-p}, \quad x - iy = \frac{1+q}{1-q}.$$

Suppose for a moment that we have $(p_1, q_1), (p_2, q_2), (p_3, q_3)$ as the (p, q) coordinates of any three points, the condition that these three points may lie in a line, is given in the form, determinant = 0, where each line of the determinant is of the form

$$1, \quad \frac{1+p}{1-p}, \quad \frac{1+q}{1-q},$$

or, what is the same thing, it is

$$1 - pq + p - q, \quad 1 - pq - p + q, \quad 1 + pq - p - q,$$

or, again

$$pq - 1, \quad p - q, \quad p + q - 2,$$

viz. the condition is

$$\begin{vmatrix} p_1 q_1 - 1, & p_1 - q_1, & p_1 + q_1 - 2 \\ p_2 q_2 - 1, & p_2 - q_2, & p_2 + q_2 - 2 \\ p_3 q_3 - 1, & p_3 - q_3, & p_3 + q_3 - 2 \end{vmatrix} = 0.$$

Suppose the (k, ϕ) coordinates of the three points are $(l, \alpha), (m, \beta), (n, \gamma)$ respectively; then this equation is

$$\begin{vmatrix} l^2 - 1, & il \sin \alpha, & l \cos \alpha - 1 \\ m^2 - 1, & im \sin \beta, & m \cos \beta - 1 \\ n^2 - 1, & in \sin \gamma, & n \cos \gamma - 1 \end{vmatrix} = 0,$$

viz. it is

$$\begin{vmatrix} l^2 - 1, & l \sin \alpha, & l \cos \alpha \\ m^2 - 1, & m \sin \beta, & m \cos \beta \\ n^2 - 1, & n \sin \gamma, & n \cos \gamma \end{vmatrix} - \begin{vmatrix} l^2 - 1, & l \sin \alpha, & 1 \\ m^2 - 1, & m \sin \beta, & 1 \\ n^2 - 1, & n \sin \gamma, & 1 \end{vmatrix} = 0,$$

or, what is the same thing, it is

$$\begin{aligned} & [(l^2 - 1)mn \sin(\beta - \gamma) + (m^2 - 1)nl \sin(\gamma - \alpha) + (n^2 - 1)lm \sin(\alpha - \beta)] \\ & + [(m^2 - n^2)l \sin \alpha + (n^2 - l^2)m \sin \beta + (l^2 - m^2)n \sin \gamma] = 0. \end{aligned}$$

If in this equation $\gamma = \pi$, and $\beta = 2\alpha$, so that $\sin(\alpha - \beta) = -\sin \alpha$, the equation will contain only terms in $\sin \alpha$ and $\sin 2\alpha$, viz. it will be

$$\begin{aligned} & [(m^2 - n^2)l + (n^2 - 1)nl - (n^2 - 1)lm] \sin \alpha \\ & + [-(l^2 - 1)mn + (n^2 - l^2)m] \sin 2\alpha = 0, \end{aligned}$$

that is,

$$m(m - 1)(n + 1)(m - n) \sin \alpha + (m^2 - 1)(n - l^2) \sin 2\alpha = 0,$$

or, what is the same thing,

$$(m + 1) \sin \alpha \{l(n + 1)(m - n) + 2m(n - l^2) \cos \alpha\} = 0,$$

which is satisfied for any values whatever of l, m, n , by a proper value of $\cos \alpha$; and is also satisfied irrespectively of the value of α if only $m = 1$ or $n = l^2$; that is, writing $m = k$ instead of l , say if $l = k$, $m = k^2$, $n = k^4$; that is, writing also ϕ in place of α , the three points

$$(k, \phi), \quad (k^2, 2\phi), \quad \text{and} \quad (k^4, \pi)$$

are in a right line; viz. the point A_1 , circle k , subtended angle ϕ ; the point A_2 , circle k^2 , subtended angle 2ϕ ; and the point V_2 , same circle, subtended angle π ; are in a right line.

The equation of the circle of unfitness can be obtained more easily in a different manner; but I have thought it worth while to give the investigation by means of the foregoing (p, q) coordinates.

Suppose that p_1, q_1 refer to the point A_1 : then we have

$$(A_1N)^2 = (x_1 - 1)^2 + y_1^2 = (x_1 + iy_1 - 1)(x_1 - iy_1 - 1) = \left(\frac{1 + p_1}{1 - p_1} - 1\right) \left(\frac{1 + q_1}{1 - q_1} - 1\right),$$

that is,

$$(A_1N)^2 = \frac{4p_1q_1}{(1 - p_1)(1 - q_1)}.$$

Similarly, if p_2, q_2 refer to the point A_2 , then

$$(A_2N)^2 = \frac{4p_2q_2}{(1 - p_2)(1 - q_2)} = \frac{4p_1^2q_1^2}{(1 - p_1^2)(1 - q_1^2)},$$

since $p_2 = p_1^2, q_2 = q_1^2$. The two are equal if

$$(1 + p_1)(1 + q_1) = p_1q_1,$$

that is,

$$p_1 + q_1 + 1 = 0.$$

Writing for a moment $x_1 + iy_1 = \xi, x_1 - iy_1 = \eta$, we have

$$p_1 = \frac{\xi - 1}{\xi + 1}, \quad q_1 = \frac{\eta - 1}{\eta + 1},$$

and the equation is

$$\frac{\xi - 1}{\xi + 1} + \frac{\eta - 1}{\eta + 1} + 1 = 0,$$

that is,

$$(\xi - 1)(\eta + 1) + (\eta - 1)(\xi + 1) + (\xi + 1)(\eta + 1) = 0,$$

or substituting for ξ, η their values, the equation is

$$3(x_1^2 + y_1^2) + 2x_1 - 1 = 0,$$

that is,

$$(x_1 + 1)(x_1 - \frac{1}{3}) + y_1^2 = 0,$$

the equation of a circle on the diameter $N'M$, which is, in fact, the before-mentioned circle of unfitness; viz. A_1 being on the circumference of this circle, or say on the boundary of the segment of unfitness, then $A_1N = A_2N$; whence also, according as A_1 is inside or outside the segment, $A_2N < A_1N$ or $> A_1N$.

Suppose A_1 to be on the circle, that is, $p_1 + q_1 + 1 = 0$; it is easy to show that the locus of A_2 is also a circle. We have in fact $(p_1 q_1)^2 + 1 = 0$, that is,

$$p_1^2 + q_1^2 + 2p_1^2 q_1^2 + 1 = 0,$$

or say

$$p_2 + q_2 + 2k^4 - 1 = 0,$$

viz. this is

$$\begin{aligned} \frac{\xi - 1}{\xi + 1} + \frac{\eta - 1}{\eta + 1} + 2k^4 - 1 &= 0, \\ (2k^4 + 1)\xi\eta + (2k^4 - 1)(\xi + \eta) + 2k^4 - 3 &= 0, \end{aligned}$$

that is,

$$x_1^2 + y_1^2 + \frac{2k^4 - 1}{2k^4 + 1} \cdot 2x_1 + \frac{2k^4 - 3}{2k^4 + 1} = 0,$$

or finally

$$(x_1 + 1) \left(x_1 - \frac{3 - 2k^4}{2k^4 + 1} \right) + y_1^2 = 0.$$

Measuring off from O in the direction of ON a distance $OS = \frac{3 - 2k^4}{2k^4 + 1}$ (always $> \frac{1}{3}$, since $k^4 < 1$), the circle in question is that on the diameter $N'S$; this is a circle touching at N' , and containing within it the circle of unfitness; if $k = 1$ (that is, for A_1 on the line QQ') it becomes identical with the circle of unfitness, but except in this limiting case it does not meet the circle of unfitness in any point on the N -side of QQ' , that is, A_1 being on the boundary of the segment of unfitness, A_2 is never on this boundary; and it thus appears that A_1 being inside the segment, A_2 is always outside the segment.

It is to be further noticed, that we have

$$(A_2N)^2 = \frac{4p_2q_2}{(1 - p_2)(1 - q_2)},$$

or

$$\frac{(A_2N)^2}{(A_1N)^2} = \frac{1 - p_1 + q_1}{1 - p_1 - q_1} = \frac{3\xi\eta + \xi + \eta - 1}{3} = \frac{3(x_1^2 + y_1^2 + \frac{2}{3}x_1 - \frac{1}{3})}{1},$$

that is,

$$\frac{(A_2N)^2}{(A_1N)^2} = 1 + \frac{3T^2}{(A_1N)^2},$$

where T is the tangential distance of A_1 from the circle of unfitness; there should, it appears to me, be some more elegant formula for the ratio $A_2N \div A_1N$ which determines whether the step is regular or irregular.

It is worth noticing how the conditions

$$\text{mod}(a - n) < \text{mod}(a + n) \quad \text{and} \quad \text{mod}(a - n) < \frac{1}{2}\text{mod}n,$$

present themselves in the real theory. Making the usual construction by means of the parabola $y = x^2$, the first condition means that the point A must be taken on the N -side of O (fig. 2); the second that, in order to the regularity of the approxi-

[Figure 2: A parabola $y = x^2$ with the axis Ox ; points N', O, A, N, A_1 shown on the axis. The construction shows that if $OA = \frac{1}{2}ON$ then $AN = NA_1$.]

mation, A must be taken at a distance from $O > \frac{1}{2}ON$; in fact, if (as in the figure) $OA = \frac{1}{2}ON$, then $AN = NA_1$, or the point A_1 is at an equal distance with A from N ; and thence, according as OA is greater or less than $\frac{1}{2}ON$, the point A_1 is nearer or further than A to or from N .

737.

ON A COVARIANT FORMULA.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. xvi. (1879), pp. 224–226.]

Starting from the equation

$$x_1 = x - \frac{fx}{f'x},$$

which presents itself in the Newton-Fourier problem, it is easy to see that, if a be a root of the equation $fx = 0$, then

$$x_1 - a = \frac{(x - a)f'x - fx}{f'x}$$

contains the factor $(x - a)^2$; that is, the equation $(x - x_1)f'x - fx = 0$, considered as an equation in x containing the parameter x_1 , will have a twofold root, if x_1 is equal to any root a of the equation $fx = 0$; and, consequently, the discriminant in regard to x of the function $(x - x_1)f'x - fx$ will contain the factor fx_1 . But if fx be of the order n , then the discriminant is of the order $2n - 2$ in x_1 , and there is consequently a remaining factor of x_1 of the order $n - 2$.

The like theorem applies to the homogeneous form

$$(\alpha y - \beta x) \left(\alpha \frac{d}{d\alpha} + \beta \frac{d}{d\beta} \right) f(\alpha, \beta) - (\alpha y - \beta x) f(\alpha, \beta),$$

which reduces itself to the foregoing on writing $\alpha = 1, \beta = 0, y = y_1 = 1$; or, changing the notation, say to the form

$$(\xi y - \eta x) \left(\xi \frac{d}{d\xi} + \eta \frac{d}{d\eta} \right) f(\xi, \eta) - (\alpha y - \beta x) f(\xi, \eta),$$

viz. the discriminant hereof in regard to ξ, η , being a function, homogeneous of the order $2n - 2$ in regard to x, y , to α, β , and to the coefficients of $f(\xi, \eta)$, will contain the factor $f(x, y)$, and there will be consequently a remaining factor of the order $n - 2$ in (x, y) , $2n - 2$ in (α, β) and $2n - 3$ in the coefficients of $f(\xi, \eta)$.

The most simple case is when $f(\xi, \eta)$ is the quadric function $(a, b, c | \xi, \eta)^2$. The form here is

$$(\xi y - \eta x) [(a\alpha + b\beta)\xi + (b\alpha + c\beta)\eta] - (\alpha y - \beta x)(a, b, c | \xi, \eta)^2 = (\mathfrak{a}, \mathfrak{b}, \mathfrak{c} | \xi, \eta)^2,$$

where the coefficients are

$$\begin{aligned} \mathfrak{a} &= y(a\alpha + b\beta) - \alpha(\alpha y - \beta x) = a\beta x + (a\alpha + 2b\beta)y, \\ \mathfrak{b} &= y(b\alpha + c\beta) - x(a\alpha + b\beta) - b(\alpha y - \beta x) = -a\alpha x + c\beta y, \\ \mathfrak{c} &= -x(b\alpha + c\beta) - c(\alpha y - \beta x) = -(2b\alpha + c\beta)x - c\alpha y; \end{aligned}$$

and we then have

$$\begin{aligned} \mathfrak{a}\mathfrak{c} - \mathfrak{b}^2 &= -(2b\alpha\beta + c\beta^2)a\alpha x^2 \\ &\quad - \{2ab\alpha^2 + (2ac + 4b^2)\alpha\beta + 2bc\beta^2\}xy - (a\alpha^2 + 2b\alpha\beta)c y^2 \\ &\quad - a\alpha\beta \cdot a\alpha x^2 - (-2ac\alpha\beta)xy - c\beta \cdot c\alpha y^2, \end{aligned}$$

which is

$$= -(a\alpha^2 + 2b\alpha\beta + c\beta^2)(ax^2 + 2bxy + cy^2).$$

The discriminant is in this case

$$= -(a, b, c | \alpha, \beta)^2 \cdot (a, b, c | x, y)^2.$$

In the case of the cubic function $(a, b, c, d \mid \xi, \eta)^3$, the form is

$$\begin{aligned} & (\xi y - \eta x) \{ 3(a\alpha + b\beta, b\alpha + c\beta, c\alpha + d\beta \mid \xi, \eta)^2 \} \\ & -(\alpha y - \beta x) (a, b, c, d \mid \xi, \eta)^3 = (\mathfrak{a}, \mathfrak{b}, \mathfrak{c}, \mathfrak{d} \mid \xi, \eta)^3, \end{aligned}$$

the values of the coefficients being

$$\begin{aligned} \mathfrak{a} &= a\beta x + (2a\alpha + 3b\beta) y, \\ \mathfrak{b} &= -a\alpha x + (b\alpha + 2c\beta) y, \\ \mathfrak{c} &= -(2b\alpha + c\beta) x + d\beta y, \\ \mathfrak{d} &= -(3c\alpha + 2d\beta) x - d\alpha y. \end{aligned}$$

Attending only to the terms in x^2 , we have

$$\begin{aligned} \mathfrak{a}\mathfrak{c} - \mathfrak{b}^2 &= -(a\alpha^2 + 2b\alpha\beta + c\beta^2) a x^2, \\ \mathfrak{a}\mathfrak{d} - \mathfrak{b}\mathfrak{c} &= -2(b\alpha^2 + 2c\alpha\beta + d\beta^2) a\alpha x^2, \\ \mathfrak{b}\mathfrak{d} - \mathfrak{c}^2 &= -\{ (3ac - 4b^2) \alpha^2 - (2ad - 4bc) \alpha\beta - c^2\beta^2 \} x^2. \end{aligned}$$

And hence, in

$$\mathfrak{a}^2\mathfrak{d}^2 + 4\mathfrak{a}\mathfrak{c}^3 + 4\mathfrak{b}^3\mathfrak{d} - 3\mathfrak{b}^2\mathfrak{c}^2 - 6\mathfrak{a}\mathfrak{b}\mathfrak{c}\mathfrak{d} = (\mathfrak{a}\mathfrak{d} - \mathfrak{b}\mathfrak{c})^2 - 4(\mathfrak{a}\mathfrak{c} - \mathfrak{b}^2)(\mathfrak{b}\mathfrak{d} - \mathfrak{c}^2),$$

we have the term

$$\begin{aligned} & 4a\alpha^2 \cdot [a(b\alpha^2 + 2c\alpha\beta + d\beta^2)^2 \\ & + (a\alpha^2 + 2b\alpha\beta + c\beta^2) \{ (3ac - 4b^2) \alpha^2 - (2ad - 4bc) \alpha\beta - c^2\beta^2 \}] x^4; \end{aligned}$$

then, forming the analogous term in y^4 , and assuming that the whole divides by $(a, b, c, d \mid x, y)^3$, and also expanding the $\alpha\beta$ -functions within the square brackets, we find

Discriminant = $4(a, b, c, d \mid x, y)^3$ multiplied by

$$\begin{aligned} & [3a^2c - 3ab^2, \\ & 2a^2d + 6abc - 8b^3, \\ & 6abd + 6ac^2 - 14b^2c, \\ & 6acd - 6bc^2, \\ & ad^2 - bc^2] \alpha^4 () / \beta^4 + [ad - b^2, \\ & 6abd - 6b^2c, \\ & 6acd + 6b^2d - 12bc^2, \\ & 2ad^2 + 6bcd - 8d^3, \\ & 3bd^2 - 3cd] \beta^4 () / \alpha^4. \end{aligned}$$

Writing down the Hessian of $(a, b, c, d \mid \alpha, \beta)^3$,

$$H = (ac - b^2, ad - bc, bd - c^2 \mid \alpha, \beta)^2,$$

and the cubicovariant

$$\Phi = (a^2d - 3abc + 2b^3, abd - 2ac^2 + b^2c, -acd + 2b^2d - bc^2, -ad^2 + 3bcd - 2c^3) (\alpha, \beta)^3.$$

It is easy to see that the coefficient of x is

$$= 3(a, b, c \mid \alpha, \beta)^2 (H - \beta\Phi);$$

hence also that of y is

$$= -3(a, b, c \mid \alpha, \beta)^2 (\beta H + \alpha\Phi), = 3(b, c, d \mid \alpha, \beta)^2 (\beta H + \alpha\Phi);$$

and the final result is that the discriminant $= 4(a, b, c, d \mid x, y)^3$ multiplied by

$$[3(a, b, c, d \mid \alpha, \beta)^2(x, y)H + (\alpha y - \beta x)\Phi].$$

It would be interesting to calculate the result for the quartic $(a, b, c, d, e \mid \xi, \eta)^4$.

March 14, 1879.

738.

NOTE ON A HYPERGEOMETRIC SERIES.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. xvi. (1879), pp. 268–270.]

In the memoir on hypergeometric series, Schwarz, “*Ueber diejenigen Fälle, &c.*,” Crelle, t. LXXV. (1873), pp. 292–335, the author shows, as part of his general theory, that the equation

$$\frac{d^2y}{dx^2} - \frac{\frac{2}{3} - \frac{1}{2}x}{x(1-x)} \cdot \frac{dy}{dx} + \frac{\frac{1}{48}}{x(1-x)^2} y = 0,$$

which belongs to the hypergeometric series $F(\frac{1}{12}, -\frac{1}{12}, \frac{2}{3}, x)$, is algebraically integrable, having in fact the two particular integrals

$$y^6 = -x + 2\sqrt{(\alpha - \alpha^5x)} \cdot \sqrt{(-\alpha^5 + \alpha x)},$$

where α is a prime sixth root of -1 , $\alpha^6 + 1 = 0$, or say $\alpha^4 - \alpha^2 + 1 = 0$ (see p. 326, α being for greater simplicity written instead of δ^2 , and the form being somewhat simplified).

It is interesting to verify this directly; writing first $y = \sqrt{Y}$ and then $x = X^2$, the equation between Y , X is easily found to be

$$\frac{d^2Y}{dX^2} - \frac{\frac{4}{3}X}{1-X^2} \frac{dY}{dX} + \frac{1}{4} \cdot \frac{1}{Y} \left(\frac{dY}{dX} \right)^2 + \frac{\frac{1}{6}}{1-X^2} Y - \frac{X}{1-X^2} Y = 0,$$

and the theorem in effect is that that equation has the two particular integrals

$$Y = \sqrt{(-P)} \pm \sqrt{(Q)},$$

P and Q being linear functions of X : in fact,

$$\begin{aligned} P &= \alpha - \alpha^5 X, \\ Q &= -\alpha^5 + \alpha X. \end{aligned}$$

Starting say from the equation

$$Y = \sqrt{(P)} + \sqrt{(Q)},$$

or, as it is convenient to write it,

$$Y = P^{\frac{1}{2}} + Q^{\frac{1}{2}},$$

where P and Q are assumed to be linear functions of X , we have

$$\begin{aligned} \frac{dY}{dX} &= \frac{1}{2} P^{-\frac{1}{2}} P' + \frac{1}{2} Q^{-\frac{1}{2}} Q', \\ \frac{d^2Y}{dX^2} &= -\frac{1}{4} P^{-\frac{3}{2}} P'^2 - \frac{1}{4} Q^{-\frac{3}{2}} Q'^2, \end{aligned}$$

and thence

$$\begin{aligned}\frac{d^2Y}{dX^2} &= -\frac{1}{4}P^{-\frac{3}{2}}P'^2 - \frac{1}{4}Q^{-\frac{3}{2}}Q'^2 - \frac{1}{4}Q^{-\frac{1}{2}}P^{-\frac{1}{2}}P'^2 - \frac{1}{4}P^{-\frac{1}{2}}Q^{-\frac{1}{2}}Q'^2, \\ Y^2 &= \frac{1}{2}(P^2 + Q^2) + P^{\frac{1}{2}}Q^{\frac{1}{2}}P' + P^{\frac{1}{2}}Q^{\frac{1}{2}}Q', \\ \left(\frac{dY}{dX}\right)^2 &= \frac{1}{4}P^{-1}P'^2 + \frac{1}{4}Q^{-1}Q'^2 + \frac{1}{2}P^{-\frac{1}{2}}Q^{-\frac{1}{2}}P'Q',\end{aligned}$$

where P', Q' are written to denote the derived functions of P, Q respectively.

Substituting these values, the resulting equation contains on the left-hand side a rational part, and a part with the factor $P^{-\frac{1}{2}}Q^{-\frac{1}{2}}$, and it is clear the equation can only be true if these two parts are separately $= 0$. We have thus two equations which ought to be verified; viz. after a slight reduction these are found to be

$$\frac{2}{3}X(P' + Q') - \frac{1}{6}(P + Q) = 0,$$

$$\frac{1}{4}PQ(QP'^2 + PQ'^2) + QP^{\frac{1}{2}}Q^{\frac{1}{2}}P'Q' + PQ^{\frac{1}{2}}P^{\frac{1}{2}}P'Q' - \frac{4}{3}X \cdot PQ(PQ' + P'Q) - \frac{1}{6} \cdot X P'Q' = 0,$$

and it is very interesting to observe the manner in which these equations are, in fact, verified by the foregoing values of P, Q .

We have

$$P + Q = (\alpha - \alpha^5)(1 + X), \quad P' + Q' = \alpha - \alpha^5,$$

and hence

$$\frac{2}{3}X(P' + Q') - \frac{1}{6}(P + Q) = -(\alpha - \alpha^5)\frac{1}{6}(1 - X),$$

or, in the first equation, the second part

$$\frac{\frac{2}{3}X}{1 - X^2}(P' + Q') - \frac{\frac{1}{6}X}{1 - X^2}(P + Q) = -(\alpha - \alpha^5)\frac{1}{1 - X^2}.$$